Advanced Computer Networks
Assignment 1-1

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Part 0: Real BGP connectivity of IIT Delhi

Information from bgp.tools[1] is used to understand the BGP connectivity of IIT Delhi. The following assumptions from [2] are adopted:

1) A fixed set of ASNs is considered to be Tier-1(an AS that has a 'N/A' upstream listed on its webpage on [1]). This pre-decided list is present on [2].
2) Any ASN that occurs in an AS path for a prefix and contains a Tier-1 AS, is considered to be an upstream provider for the final destination(originating AS) in the path.

All Tier-1 ISPs are peered with each other for route announcements and route propagation.

IIT Delhi forms an independent AS with an ID of **132780**. It announces four IPv4 prefixes : **103.27.8.0/24, 103.27.9.0/24, 103.27.10.0/24 and 103.27.11.0/24**[3].

IIT Delhi has a single upstream relationship with **AS55824** (NKN Core Network)[4] that propagates IIT Delhi's prefixes to its peers. AS55824(NKN Core Network) forms the only real-world BGP peer for IIT Delhi[5]. The route propagation and upstream relationships starting from IITD AS can be observed in Figure 1.

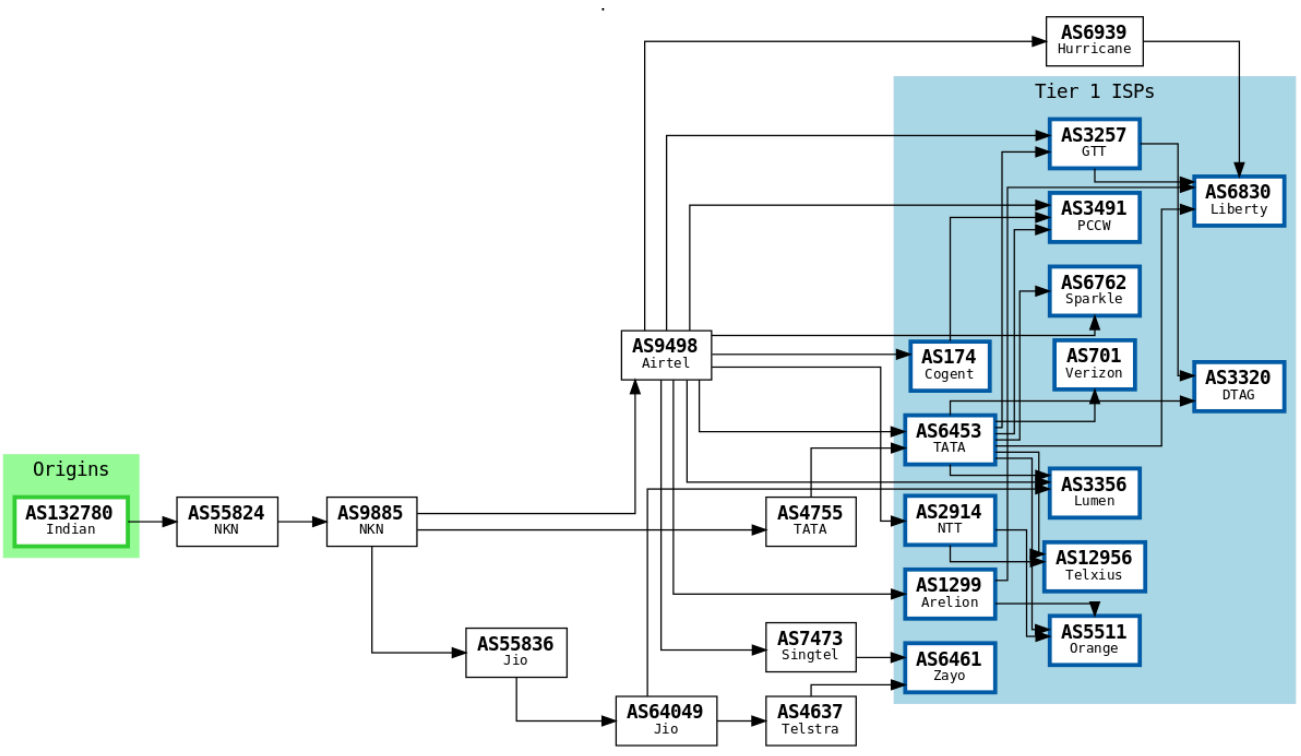


Figure 1: Route propagation path for prefixes from IIT Delhi to reach Tier-1 ISPs(marked in blue)[6]

Based on Figure 1, the second assumption listed above, and by recursively following the upstream links on bgp.tools site[1] starting from IITD AS[4], the following upstream relationships are observed until any Tier-1 ISP is reached.

Sno	AS	Name	Upstream provider ASN	Comments
1	132780	IIT Delhi	55824	Originating AS
2	55824	NKN Core Network	9885	Not a traditional ISP
3	9885	NKN INTERNET GATEWAY NETWORK	9498, 4755, 55836	Not a traditional ISP
4	9498	Bharti Airtel Ltd.	6453 and 24 more[7]	Traditional ISP. Has a Tier-1 upstream
5	4755	TATA Communications	6453 and 5 more[8]	Traditional ISP. Has a Tier-1 upstream
6	55836	Reliance Jio Infocomm Limited	64049 and 8 more[9]	Traditional ISP
7	6453	TATA Communications (America) Inc	N/A	Tier-1 ISP
8	64049	Reliance Jio Infocomm Pte. Ltd.	3356 and 7 more[10]	Traditional ISP. Has a Tier-1 upstream
9	3356	Lumen (Level 3)	N/A	Tier-1 ISP

Table 1: List of all upstream AS starting from IIT Delhi, ending with the first Tier-1 ISP reached in a path

In the table above, all AS(Sno 2 - 9), except Sno 1(IITD itself), form the upstream ISPs that announce IIT Delhi's routes to the global Internet. The bgp.tools website observes these upstream relationships by monitoring AS paths received by its Route collectors placed at different parts of the world. Here, IIT Delhi has three traditional upstream ISPs that charge for Internet access: **AS9498(Bharti Airtel Ltd.), AS4755(TATA Communications), and AS55836(Reliance Jio Infocomm Limited)**. The financial relationship between these ISPs(Tata, Reliance, and Airtel), NKN, and IIT Delhi is unknown and outside the scope of this report.

Part A: Baseline

In the following console excerpt figures from NS3[11] runs, the first entry in a line represents simulated time, the second is the router/node ID, and the remaining is the message with the LOGIC prefix_level present on the IPv4 L3 layer. Part A is run using the following command after compiling bgp.cc:

```
NS_LOG="*=prefix_time|prefix_node|prefix_level:Ipv4L3Protocol=level_logic|prefix_time|prefix_node|prefix_level" ./build/scratch/ns3.45-bgp-default
```

```
+2.000000000s 0 Ipv4L3Protocol:Send(0x14570d360, 0x600001a3d5e0, 10.0.1.1, 10.0.2.2, 17, 0x600003f6c360)
+2.000000000s 0 Ipv4L3Protocol:BuildHeader(0x14570d360, 10.0.1.1, 10.0.2.2, 17, 1032, 64, 0, 1)
+2.000000000s 0 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
+2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 127.255.255.255
+2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 10.0.1.255
+2.000000000s 0 Ipv4L3Protocol:SendRealOut(0x14570d360, 0x600003f6c360, 0x600001a3d730, 0x16bdfa380)
+2.000000000s 0 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.000000000s 0 [LOGIC] Send to gateway 10.0.1.2
+2.000000000s 0 [INFO ] At time +2s client sent 1024 bytes to 10.0.2.2 port 9
+2.002843200s 1 Ipv4L3Protocol:Receive(0x145606110, 0x145606a70, 0x600001a3d7a0, 2048,
02-06-00:00:00:00:00:01, 02-06-00:00:00:00:00:02, 0)
+2.002843200s 1 [LOGIC] Packet from 02-06-00:00:00:00:00:01 received on node 1
+2.002843200s 1 [LOGIC] Forwarding state: true
+2.002843200s 1 [LOGIC] Forwarding state: true
+2.002843200s 1 Ipv4L3Protocol:SourceAddressSelection(0x145606110, 2, " ", 10.0.2.0)
+2.002843200s 1 Ipv4L3Protocol:IpForward(0x145606110, 0x600003f6f2a0, 0x600001a3d570, tos 0x0 DSCP
Default ECN Not-ECT ttl 64 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.2.
2)
+2.002843200s 1 [LOGIC] Forwarding logic for node: 1
+2.002843200s 1 Ipv4L3Protocol:SendRealOut(0x145606110, 0x600003f6f2a0, 0x600001a3d5e0, 0x16bdf9dc0)
+2.002843200s 1 [LOGIC] Send via NetDevice ifIndex 2 ipv4InterfaceIndex 2
+2.002843200s 1 [LOGIC] Send to destination 10.0.2.2
+2.005686400s 2 Ipv4L3Protocol:Receive(0x145606490, 0x145606e50, 0x600001a3d810, 2048,
02-06-00:00:00:00:00:03, 02-06-00:00:00:00:00:04, 0)
+2.005686400s 2 [LOGIC] Packet from 02-06-00:00:00:00:00:03 received on node 2
+2.005686400s 2 [LOGIC] For me (destination 10.0.2.2 match)
+2.005686400s 2 Ipv4L3Protocol:LocalDeliver(0x145606490, 0x600001a3d7a0, 0x16bdfa210, 1)
+2.005686400s 2 [INFO ] At time +2.00569s server received 1024 bytes from 10.0.1.1 port 49153
+2.005686400s 2 Ipv4L3Protocol:Send(0x145606490, 0x600001a3d960, 10.0.2.2, 10.0.1.1, 17, 0x600003f6f780)
+2.005686400s 2 Ipv4L3Protocol:BuildHeader(0x145606490, 10.0.2.2, 10.0.1.1, 17, 1032, 64, 0, 1)
+2.005686400s 2 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
+2.005686400s 2 [LOGIC] Testing address 10.0.1.1 with subnet-directed broadcast 127.255.255.255
+2.005686400s 2 [LOGIC] Testing address 10.0.1.1 with subnet-directed broadcast 10.0.2.255
+2.005686400s 2 Ipv4L3Protocol:SendRealOut(0x145606490, 0x600003f6f780, 0x600001a3dab0, 0x16bdf9220)
+2.005686400s 2 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.005686400s 2 [LOGIC] Send to gateway 10.0.2.1
+2.005686400s 2 [INFO ] At time +2.00569s server sent 1024 bytes to 10.0.1.1 port 49153
+2.008529600s 1 Ipv4L3Protocol:Receive(0x145606110, 0x145606c60, 0x600001a3db20, 2048,
02-06-00:00:00:00:00:04, 02-06-00:00:00:00:00:03, 0)
+2.008529600s 1 [LOGIC] Packet from 02-06-00:00:00:00:00:04 received on node 1
+2.008529600s 1 [LOGIC] Forwarding state: true
+2.008529600s 1 [LOGIC] Forwarding state: true
+2.008529600s 1 Ipv4L3Protocol:SourceAddressSelection(0x145606110, 1, " ", 10.0.1.0)
+2.008529600s 1 Ipv4L3Protocol:IpForward(0x145606110, 0x600003f6fd50, 0x600001a3d810, tos 0x0 DSCP
Default ECN Not-ECT ttl 64 id 0 protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.1.
1)
+2.008529600s 1 [LOGIC] Forwarding logic for node: 1
+2.008529600s 1 Ipv4L3Protocol:SendRealOut(0x145606110, 0x600003f6fd50, 0x600001a3d5e0, 0x16bdf9dc0)
+2.008529600s 1 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.008529600s 1 [LOGIC] Send to destination 10.0.1.1
+2.011372800s 0 Ipv4L3Protocol:Receive(0x14570d360, 0x145606880, 0x600001a3d730, 2048,
02-06-00:00:00:00:00:02, 02-06-00:00:00:00:00:01, 0)
+2.011372800s 0 [LOGIC] Packet from 02-06-00:00:00:00:00:02 received on node 0
+2.011372800s 0 [LOGIC] For me (destination 10.0.1.1 match)
+2.011372800s 0 Ipv4L3Protocol:LocalDeliver(0x14570d360, 0x600001a3db20, 0x16bdfa210, 1)
+2.011372800s 0 [INFO ] At time +2.01137s client received 1024 bytes from 10.0.2.2 port 9
```

Figure 2: Console log excerpt from BGP baseline run taken from bgplite.log

In Figure 2, the client sends a message from Router 0(AS1) as highlighted in line 112, which is forwarded to Router 1(AS2) on 10.0.1.2 as per line 111. Router 1 receives the message and forwards this to Router 2(AS3) on 10.0.2.2 as per line 122. The server on Router 2 receives this message as per line 127. The server on Router 2 sends the response back as per line 136. Router 1 again receives the message and forwards it to Router 0 on 10.0.1.1 as per line 146. Finally, the client receives the server response as per line 151

Part B

A direct link between AS1 and AS3 is added, parallel to the AS1–AS2–AS3 path, with higher bandwidth(20 Mbps) and reduced latency(1ms). Part B is run using the following command after compiling bgp_partB.cc

```
NS_LOG="*=prefix_time|prefix_node|prefix_level:Ipv4L3Protocol=level_logic|pr
fix_time|prefix_node|prefix_level" ./build/scratch/ns3.45-bgp_partB-default
```

```
116  +2.000000000s 0 Ipv4L3Protocol:Send(0x12a814bc0, 0x600000cb4930, 10.0.3.1, 10.0.2.2, 17,
     0x6000029e3510)
117  +2.000000000s 0 Ipv4L3Protocol:BuildHeader(0x12a814bc0, 10.0.3.1, 10.0.2.2, 17, 1032, 64, 0, 1)
118  +2.000000000s 0 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
119  +2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 127.255.255.255
120  +2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 10.0.1.255
121  +2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 10.0.3.255
122  +2.000000000s 0 Ipv4L3Protocol:SendRealOut(0x12a814bc0, 0x6000029e3510, 0x600000cb9500,
     0x16b15e210)
123  +2.000000000s 0 [LOGIC] Send via NetDevice ifIndex 2 ipv4InterfaceIndex 2
124  +2.000000000s 0 [LOGIC] Send to gateway 10.0.3.2
125  +2.000000000s 0 [INFO ] At time +2s client sent 1024 bytes to 10.0.2.2 port 9
126  +2.001421600s 2 Ipv4L3Protocol:Receive(0x128e04790, 0x128f05460, 0x600000cb9490, 2048,
     02-06-00:00:00:00:00:05, 02-06-00:00:00:00:00:06, 0)
127  +2.001421600s 2 [LOGIC] Packet from 02-06-00:00:00:00:00:05 received on node 2
128  +2.001421600s 2 [LOGIC] For me (destination 10.0.2.2 match) on another interface
129  +2.001421600s 2 Ipv4L3Protocol:LocalDeliver(0x128e04790, 0x600000cb9420, 0x16b15e0a0, 2)
130  +2.001421600s 2 [INFO ] At time +2.00142s server received 1024 bytes from 10.0.3.1 port 49153
131  +2.001421600s 2 Ipv4L3Protocol:SourceAddressSelection(0x128e04790, 2, " ", 10.0.3.0)
132  +2.001421600s 2 Ipv4L3Protocol:Send(0x128e04790, 0x600000cb91f0, 10.0.3.2, 10.0.3.1, 17,
     0x6000029ddce0)
133  +2.001421600s 2 Ipv4L3Protocol:BuildHeader(0x128e04790, 10.0.3.2, 10.0.3.1, 17, 1032, 64, 0, 1)
134  +2.001421600s 2 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
135  +2.001421600s 2 [LOGIC] Testing address 10.0.3.1 with subnet-directed broadcast 127.255.255.255
136  +2.001421600s 2 [LOGIC] Testing address 10.0.3.1 with subnet-directed broadcast 10.0.2.255
137  +2.001421600s 2 [LOGIC] Testing address 10.0.3.1 with subnet-directed broadcast 10.0.3.255
138  +2.001421600s 2 Ipv4L3Protocol:SendRealOut(0x128e04790, 0x6000029ddce0, 0x600000cb0460,
     0x16b15d0b0)
139  +2.001421600s 2 [LOGIC] Send via NetDevice ifIndex 2 ipv4InterfaceIndex 2
140  +2.001421600s 2 [LOGIC] Send to destination 10.0.3.1
141  +2.001421600s 2 [INFO ] At time +2.00142s server sent 1024 bytes to 10.0.3.1 port 49153
142  +2.002843200s 0 Ipv4L3Protocol:Receive(0x12a814bc0, 0x128f05120, 0x600000cb04d0, 2048,
     02-06-00:00:00:00:00:06, 02-06-00:00:00:00:00:05, 0)
143  +2.002843200s 0 [LOGIC] Packet from 02-06-00:00:00:00:00:06 received on node 0
144  +2.002843200s 0 [LOGIC] For me (destination 10.0.3.1 match)
145  +2.002843200s 0 Ipv4L3Protocol:LocalDeliver(0x12a814bc0, 0x600000cb0380, 0x16b15e0a0, 2)
146  +2.002843200s 0 [INFO ] At time +2.00284s client received 1024 bytes from 10.0.3.2 port 9
```

Figure 3: Console log excerpt from BGP Part B run taken from bgp_partB.log

In Figure 3, the client on Router 0(AS1) sends the message directly to the server on Router 2(AS3) using the new IP of the added link i.e 10.0.3.2, as per lines 124 and 125. The server on Router 2 receives the message as per line 130. It sends a response back ot the client directly on 10.0.3.1 as per lines 140 and 141. The client on Router 0 receives the response as per line 146. Router 1(AS1) is completely skipped, which can be verified from its absence in the log.

Part C

Building on top of part B, a static route is added to Router 0(AS1) for forward path, and Router 2(AS3) for the return path, so that traffic is forcefully forwarded to Router 1(AS2) despite there being a direct path between AS1 and AS3 available. The code additions for adding a static route on AS1 and AS3, both pointing to AS2, are given below:

```
Ipv4StaticRoutingHelper ipv4RoutingHelper;

Ptr<Ipv4StaticRouting> router0_route =
ipv4RoutingHelper.GetStaticRouting(routers.Get(0)->GetObject<Ipv4> ());

router0_route->AddHostRouteTo(Ipv4Address("10.0.2.2"),
Ipv4Address("10.0.1.2"), 1);

Ptr<Ipv4StaticRouting> router2_route =
ipv4RoutingHelper.GetStaticRouting(routers.Get(2)->GetObject<Ipv4> ());

router2_route->AddHostRouteTo(Ipv4Address("10.0.1.1"),
Ipv4Address("10.0.2.1"), 1);
```

Part C is run using the following command after compiling bgp_partC.cc

```
NS_LOG="*=prefix_time|prefix_node|prefix_level:Ipv4L3Protocol=level_logic|pr
fix_time|prefix_node|prefix_level" ./build/scratch/ns3.45-bgp_partC-default
```

```
+2.000000000s 0 Ipv4L3Protocol:SourceAddressSelection(0x146f0c8a0, 1, " ", 10.0.2.2)
+2.000000000s 0 Ipv4L3Protocol:Send(0x146f0c8a0, 0x600001fc1810, 10.0.1.1, 10.0.2.2, 17, 0x600003aacf90)
+2.000000000s 0 Ipv4L3Protocol:BuildHeader(0x146f0c8a0, 10.0.1.1, 10.0.2.2, 17, 1032, 64, 0, 1)
+2.000000000s 0 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
+2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 127.255.255.255
+2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 10.0.1.255
+2.000000000s 0 [LOGIC] Testing address 10.0.2.2 with subnet-directed broadcast 10.0.3.255
+2.000000000s 0 Ipv4L3Protocol:SendRealOut(0x146f0c8a0, 0x600003aacf90, 0x600001fc1960, 0x16b6ce200)
+2.000000000s 0 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.000000000s 0 [LOGIC] Send to gateway 10.0.1.2
+2.000000000s 0 [INFO ] At time +2s client sent 1024 bytes to 10.0.2.2 port 9
+2.001421600s 1 Ipv4L3Protocol:Receive(0x146e04980, 0x146e056e0, 0x600001fc19d0, 2048, 02-06-00:00:00:00:00:01, 02-06-00:00:00:00:00:02, 0)
+2.001421600s 1 [LOGIC] Packet from 02-06-00:00:00:00:00:01 received on node 1
+2.001421600s 1 [LOGIC] Forwarding state: true
+2.001421600s 1 [LOGIC] Forwarding state: true
+2.001421600s 1 Ipv4L3Protocol:SourceAddressSelection(0x146e04980, 2, " ", 10.0.2.0)
+2.001421600s 1 Ipv4L3Protocol:IpForward(0x146e04980, 0x600003aaf7e0, 0x600001fc17a0, tos 0x0 DSCP Default ECN Not-ECT ttl 64 id 0
protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.1.1 > 10.0.2.2)
+2.001421600s 1 [LOGIC] Forwarding logic for node: 1
+2.001421600s 1 Ipv4L3Protocol:SendRealOut(0x146e04980, 0x600003aaf7e0, 0x600001fc1810, 0x16b6cdc40)
+2.001421600s 1 [LOGIC] Send via NetDevice ifIndex 2 ipv4InterfaceIndex 2
+2.001421600s 1 [LOGIC] Send to destination 10.0.2.2
+2.002843200s 2 Ipv4L3Protocol:Receive(0x146e04e90, 0x146e05d60, 0x600001fc1a40, 2048, 02-06-00:00:00:00:00:03, 02-06-00:00:00:00:00:04, 0)
+2.002843200s 2 [LOGIC] Packet from 02-06-00:00:00:00:00:03 received on node 2
+2.002843200s 2 [LOGIC] For me (destination 10.0.2.2 match)
+2.002843200s 2 Ipv4L3Protocol:LocalDeliver(0x146e04e90, 0x600001fc19d0, 0x16b6ce090, 1)
+2.002843200s 2 [INFO ] At time +2.00284s server received 1024 bytes from 10.0.1.1 port 49153
+2.002843200s 2 Ipv4L3Protocol:SourceAddressSelection(0x146e04e90, 1, " ", 10.0.1.1)
+2.002843200s 2 Ipv4L3Protocol:Send(0x146e04e90, 0x600001fc1b90, 10.0.2.2, 10.0.1.1, 17, 0x600003aafcc0)
+2.002843200s 2 Ipv4L3Protocol:BuildHeader(0x146e04e90, 10.0.2.2, 10.0.1.1, 17, 1032, 64, 0, 1)
+2.002843200s 2 [LOGIC] Ipv4L3Protocol::Send case 1b:  passed in with route and valid gateway
+2.002843200s 2 [LOGIC] Testing address 10.0.1.1 with subnet-directed broadcast 127.255.255.255
+2.002843200s 2 [LOGIC] Testing address 10.0.1.1 with subnet-directed broadcast 10.0.2.255
+2.002843200s 2 [LOGIC] Testing address 10.0.1.1 with subnet-directed broadcast 10.0.3.255
+2.002843200s 2 Ipv4L3Protocol:SendRealOut(0x146e04e90, 0x600003aafcc0, 0x600001fc1ce0, 0x16b6cd0a0)
+2.002843200s 2 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.002843200s 2 [LOGIC] Send to gateway 10.0.2.1
+2.002843200s 2 [INFO ] At time +2.00284s server sent 1024 bytes to 10.0.1.1 port 49153
+2.004264800s 1 Ipv4L3Protocol:Receive(0x146e04980, 0x146e05a20, 0x600001fc1d50, 2048, 02-06-00:00:00:00:00:04, 02-06-00:00:00:00:00:03, 0)
+2.004264800s 1 [LOGIC] Packet from 02-06-00:00:00:00:00:04 received on node 1
+2.004264800s 1 [LOGIC] Forwarding state: true
+2.004264800s 1 [LOGIC] Forwarding state: true
+2.004264800s 1 Ipv4L3Protocol:SourceAddressSelection(0x146e04980, 1, " ", 10.0.1.0)
+2.004264800s 1 Ipv4L3Protocol:IpForward(0x146e04980, 0x600003a92d00, 0x600001fc1a40, tos 0x0 DSCP Default ECN Not-ECT ttl 64 id 0
protocol 17 offset (bytes) 0 flags [none] length: 1052 10.0.2.2 > 10.0.1.1)
+2.004264800s 1 [LOGIC] Forwarding logic for node: 1
+2.004264800s 1 Ipv4L3Protocol:SendRealOut(0x146e04980, 0x600003a92d00, 0x600001fc1810, 0x16b6cdc40)
+2.004264800s 1 [LOGIC] Send via NetDevice ifIndex 1 ipv4InterfaceIndex 1
+2.004264800s 1 [LOGIC] Send to destination 10.0.1.1
+2.005686400s 0 Ipv4L3Protocol:Receive(0x146f0c8a0, 0x146e053a0, 0x600001fc1960, 2048, 02-06-00:00:00:00:00:02, 02-06-00:00:00:00:00:01, 0)
+2.005686400s 0 [LOGIC] Packet from 02-06-00:00:00:00:00:02 received on node 0
+2.005686400s 0 [LOGIC] For me (destination 10.0.1.1 match)
+2.005686400s 0 Ipv4L3Protocol:LocalDeliver(0x146f0c8a0, 0x600001fc1d50, 0x16b6ce090, 1)
+2.005686400s 0 [INFO ] At time +2.00569s client received 1024 bytes from 10.0.2.2 port 9
```

Figure 4: Console log excerpt from BGP Part C run taken from bgp_partC.log

In Figure 4, the same end-to-end behavior as part A is repeated. Due to the existence of a static route, the packet sent by the client now travels from Router 0(AS1) to Router 1(AS2) to Router2(AS3) in the forward path, and traverses the same path in reverse for the response packet from the server.

References

1) https://bgp.tools/
2) https://bgp.tools/kb/what-is-a-upstream
3) https://bgp.tools/as/132780#prefixes
4) https://bgp.tools/as/132780#upstreams
5) https://bgp.tools/as/132780#peers
6) https://bgp.tools/as/132780#connectivity
7) https://bgp.tools/as/9498#upstreams
8) https://bgp.tools/as/4755#upstreams
9) https://bgp.tools/as/55836#upstreams
10) https://bgp.tools/as/64049#upstreams
11) https://www.nsnam.org/